

Physics A

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Duration: 4 hours. The use of calculators is not allowed for this paper.

English translation

Unofficial English translation of the original French paper (`physique-a-pc_paper_fr.pdf`, in this folder). Question numbers, notation and figures are unchanged. In case of doubt, the French original prevails.

Modern techniques for the mass spectrometry of complex molecules

The two parts are independent.

Most of the questions can be answered concisely.

Identifying complex molecules is an important challenge in chemistry and biology. It often relies on a precise measurement of the mass of the molecule, from which its molecular formula can be deduced. The measurement is carried out in two steps. First, one or more electrons are removed from the molecule. Second, the trajectory of the ion thus formed in an electromagnetic field is studied by means of a mass spectrometer.

Various techniques are used for each of these two steps. In the first part, we study electrospray ionisation, known by its acronym ESI, which earned its inventor, the American John Fenn, the Nobel Prize in Chemistry in 2002. This technique ionises very large molecules without breaking them, and naturally produces highly charged ions, which guarantee a more precise measurement. In the second part, we study so-called “ion cyclotron resonance” mass spectrometry, known by its acronym FT-ICR-MS, which is very precise and can analyse a large number of compounds present in a sample simultaneously.

I – Electrospray ionisation

Electrospray is an electrostatic method. A high potential difference (a few kV) is applied between the end of a small steel tube, called a capillary tube, and an electrode located near the outlet of the tube. A conducting liquid containing the substance under study flows very slowly through the tube. The electric field pulls on the surface of the liquid leaving the tube, which then forms an almost perfect, stationary cone (figure 1). A thin jet leaves the tip of this liquid cone and then breaks up into microscopic charged droplets. The evaporation of these droplets leads to isolated ions, which are sent into the spectrometer.

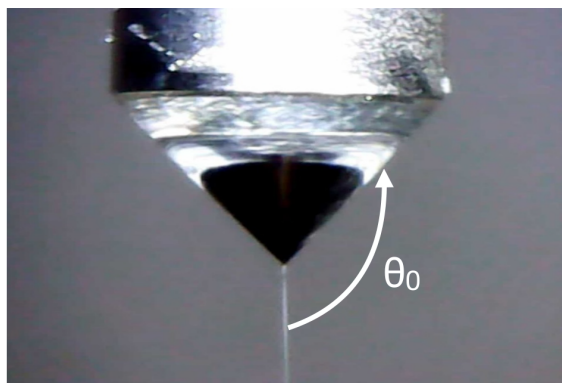


FIGURE 1 – An electrospray source showing, from top to bottom: the capillary tube inside which the liquid under study flows (diameter 0.2 mm); the cone formed by the liquid leaving the tube under the effect of the electrostatic force; and the thin jet coming from the tip of the cone. The break-up of this jet produces the ions that will be sent into the mass spectrometer. [Illustration adapted from <https://sprucescience.com/products/taylor-cone/>]

Parallel-plate capacitor

We first derive the expression of the electrostatic force exerted on a plate of a parallel-plate capacitor.

1. Recall the relation between the surface charge density on one of the plates, denoted σ , and the electric field E between the plates.
2. Recall the expression of the electrostatic energy per unit volume in terms of E .
3. Deduce the force per unit area, denoted P , exerted on one of the plates of the capacitor, and give its direction. Express P in terms of E , then in terms of σ .

We shall accept that the expressions of P thus obtained remain valid at every point of the surface of a conductor of arbitrary shape in electrostatics: P is the force per unit area experienced by the surface of the conductor.

Charged spherical drop; Rayleigh stability criterion

We now study the effect of electrostatic forces on a spherical conducting drop of radius R , carrying a charge Q , subject to no external force.

4. Determine the electric field outside the drop, and deduce the expression of P .
5. We accept that a cohesive force also acts at every point of the surface. It arises from the interactions between the molecules of the drop and is called the surface-tension force. It is directed inwards and its magnitude per unit area is γ/R , where γ is a constant coefficient. Show that if Q exceeds a limiting value, to be expressed in terms of R and γ , the electrostatic force overcomes this cohesive force.¹ The drop then breaks up, either by emitting an ion or into smaller drops: this phenomenon is called the Rayleigh instability.

Taylor cone

Finally, we study the formation of the cone shown in figure 1.²

6. The flow rate is assumed low enough for currents to be neglected, and we work within the electrostatic approximation. What equation must the electrostatic potential $V(\vec{r})$ satisfy in the region outside the cone, assumed to be free of charge? What condition must it satisfy on the surface of the cone, which consists of a conducting liquid?

We choose spherical coordinates (r, θ, ϕ) centred on the apex of the cone, such that the direction of the jet corresponds to $\theta = 0$. In this coordinate system, the equation of the surface of the cone is $\theta = \theta_0$, with $\theta_0 > \frac{\pi}{2}$ (figure 1).

7. We accept that a surface-tension force, directed inwards and proportional to γ/r , acts at every point of the surface of the cone. For the cone to be in equilibrium, the electrostatic force must exactly balance the surface-tension force at every point. How must the magnitude of the electric field vary with r for this to hold?

8. We look for a solution of the equation that determines the electrostatic potential, of the form $V(r, \theta, \phi) = V_0 + r^\delta f(\theta)$, where V_0 and δ are constants. What value of δ is compatible with the result of question 7?

¹A more careful model, worked out in 1879 by the British physicist Rayleigh (1842–1919), gives a limiting value larger by a factor $\sqrt{2}$.

²This phenomenon was modelled in 1964 by the British physicist Taylor (1886–1975).

- 9.** Explain why the equation obtained in question 6 reduces to a second-order differential equation for $f(\theta)$. You need not write it out.
- 10.** What value of $f(\theta_0)$ is compatible with the condition obtained in question 6? Solving the equation numerically with this condition gives an apex angle of the cone of about 49.3° .
- 11.** How does the charge density on the surface of the cone vary with r ? Why is it plausible that the jet leaving the tip of the cone gives rise to highly charged droplets? It is these droplets that, through the Rayleigh instability, give rise to the multiply charged ions that are then analysed in the mass spectrometer.

II – Ion cyclotron resonance mass spectrometry

12. Several techniques for measuring the mass of an ion consist in measuring its motion in a known electromagnetic field. Explain why this motion does not give direct access to m , but to the ratio m/q , where q is the electric charge of the ion.

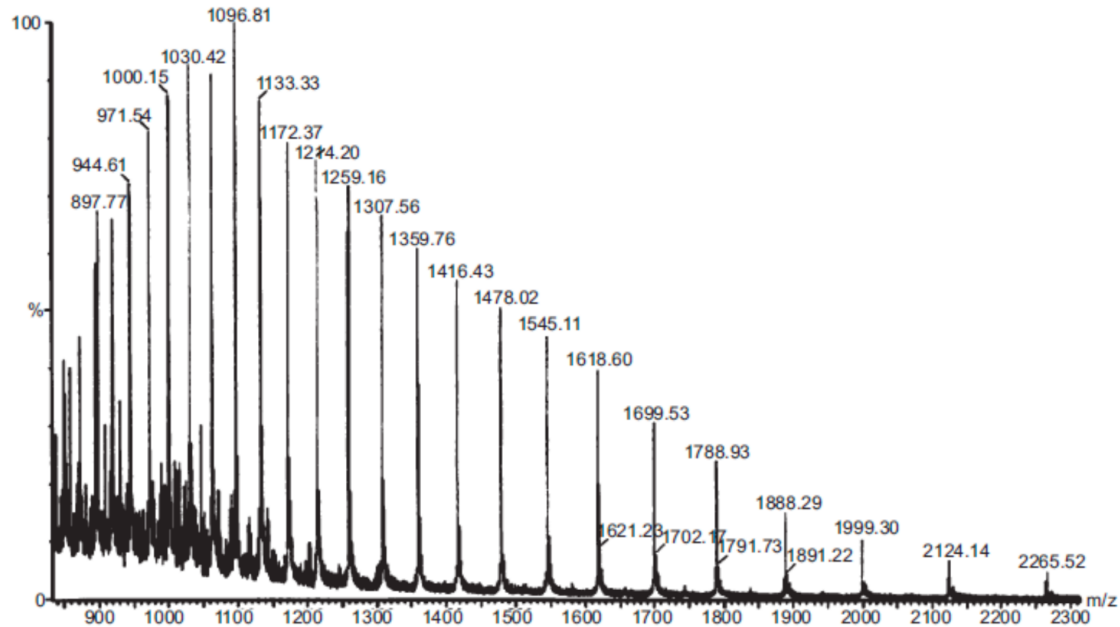


FIGURE 2 – Mass spectrum of a protein measured in a spectrometer: the detected intensity, proportional to the number of ions, as a function of the ratio m/z of the ions. The mass m is in atomic mass units, which means that it is approximately equal to the total number of neutrons and protons contained in the protein. [*Principles and Applications of Clinical Mass Spectrometry*, eds. Nader Rifai, Andrea Rita Horvath and Carl Wittwer, Elsevier, 2018.]

13. Recall why the charge of the ion is of the form $q = ze$, where e is the elementary charge and z a positive integer. Figure 2 shows the distribution of m/z in a sample analysed by a mass spectrometer. The main peaks correspond to different degrees of ionisation z of the same protein. Deduce from these data the mass m of the protein. You may begin by determining the values of z corresponding to the peaks located at $m/z = 1000.15$ and $m/z = 1999.30$. Estimate the order of magnitude of the number of atoms contained in the protein.

We now study in detail the technique of ion cyclotron resonance mass spectrometry. It sets in motion only the ions for which m/z has a precise value, by means of a constant magnetic field and an electric field oscillating at an angular frequency ω .

14. The ions produced are sent into an analysis cell in which there is a uniform, constant magnetic field $\vec{B} = B\vec{e}_z$, where $\vec{e}_z$ is the unit vector along the z axis and $B > 0$. Describe a simple experimental device that creates such a magnetic field.

15. Write the equation of motion of the ion under the action of this magnetic field, in Cartesian coordinates $(x(t), y(t), z(t))$.

16. We introduce the complex coordinate $Z(t) = x(t) + iy(t)$. Write the equation of motion for $Z(t)$, and look for a particular solution of the form $Z(t) = Z_0 \exp(-i\omega_0 t)$ with Z_0 and ω_0 non-zero. Express ω_0 in terms of q , B and m . Check your result carefully: it will be used in the rest of the problem.

Excitation

- 17.** By means of an electric field, a time-dependent excitation force perpendicular to the magnetic field is applied to the ion. Its components are $F_x(t) = F \cos \omega t$ and $F_y(t) = -F \sin \omega t$. Rewrite the equation of motion for $Z(t)$ taking this forcing into account.
- 18.** We assume that the ion, of velocity $\vec{v}$, experiences a friction force $-(m/\tau)\vec{v}$ in addition to the magnetic force and the excitation force. This force models its interaction with the gas contained in the cell. What is the dimension of τ ? In what context have you come across a similar friction force?
- 19.** Write the equation of motion for $Z(t)$ under the combined action of the three forces. Look for a steady-state (forced) solution of the form $Z(t) = Z_0 \exp(-i\omega t)$, and express the amplitude Z_0 in terms of F , m , ω , ω_0 and τ .
- 20.** Show that if the friction force is weak enough, the amplitude $Z_0(\omega)$ exhibits a resonance at the angular frequency ω_0 . Determine the order of magnitude of the width $\Delta\omega$ of the resonance in terms of τ .
- 21.** The ratio m/z is deduced from a measurement of the resonance angular frequency ω_0 . To distinguish molecules with very close masses, why is it important for z and the magnetic field B to be as large as possible?
- 22.** In practice, a uniform electric field along the x axis is used, $E_x(t) = E \cos \omega t$. Show that the excitation force it exerts on the ion can be written as the sum of two components with angular frequencies ω and $-\omega$, each of the form of question 17. Using the previous results, deduce the motion of the ion in the forced regime. Which of the two components causes the resonance?
- 23.** Describe a simple experimental device that creates such an electric field.
- 24.** We now work exactly at resonance, $\omega = \omega_0$. The friction force is neglected. Of the two components of the electric field identified in question 22, only the one that creates the resonance is taken into account. Write the equation of motion for $Z(t)$.
- 25.** Look for a solution of the form $Z(t) = iA t \exp(-i\omega_0 t)$, where A is a constant to be expressed in terms of E and B .
- 26.** Sketch the trajectory followed by the ion in the (x, y) plane. You may first write its equation in polar coordinates in the form $r = f(\theta)$. Describe the respective roles of the electric and magnetic fields in the motion.
- 27.** We shall see below that the ions are easier to detect at resonance when the amplitude of their motion is larger. On the other hand, we want to prevent them from leaving the analysis cell, whose characteristic size d is a few centimetres. Estimate, as a literal order of magnitude, the optimal value of the time t_E during which the electric field should be applied, in terms of d , E and B .
- 28.** We have described a device in which the electric field and the magnetic field are both uniform. Of these two uniformity conditions, which one seems to you the more important?
- 29.** The ions entering the analysis cell have a thermal motion before being accelerated by the electric field. Estimate the literal order of magnitude of the corresponding momentum in terms of m and the temperature T .
- 30.** Deduce the order of magnitude of the radius r of the trajectory in the plane perpendicular to the magnetic field $\vec{B}$. Excitation by the electric field is effective only if this initial radius r is much smaller than the characteristic size d of the cell. Show that this condition holds only if the mass m is much smaller than a critical mass $m_{\max}$, to be expressed in terms of q , B , d and T .

Trapping

- 31.** Explain briefly why, with the device described above, the ions can escape in the z direction.
- 32.** In addition to the fields already mentioned, electrodes create an electrostatic potential of the form $V(x, y, z) = \frac{\alpha}{2}(-x^2 - y^2 + \beta z^2)$ inside the analysis cell, with $\alpha > 0$. Write the components of the electric field $\vec{E}$ corresponding to this so-called “trapping” potential. The cell is assumed to be free of charge. Determine the value of β .
- We now study the effect of the trapping potential on the motion of the ions. In questions 33 to 36, neither the excitation force nor the friction force is taken into account.
- 33.** Show that the ion oscillates along the z direction about the origin at an angular frequency ω_p , to be expressed in terms of q , m and α .
- 34.** Establish how the equation of motion for $Z(t) = x(t) + iy(t)$ obtained in question 16 is modified in the presence of the potential $V(x, y, z)$.
- 35.** Look for particular solutions of the form $Z(t) = Z_0 \exp(-i\omega t)$ with Z_0 non-zero. The trapping field is assumed weak enough for the condition $\omega_p \ll \omega_0$ to hold. Show that there are then two real solutions for ω , denoted ω_+ and ω_- , with $\omega_+ > \omega_-$, and simplify their expressions in the limit $\omega_p \ll \omega_0$.
- 36.** Show that the trapping field leads to a small change in the resonance angular frequency. Show also that a new, much slower oscillatory motion appears in the plane perpendicular to the magnetic field, and that its frequency is independent of the ratio m/q .

Detection

After an electric excitation force has been applied for a time t_E , we want to measure the motion of the ions. To do so, two detection electrodes are placed at $y = -d/2$ and $y = d/2$. They are identical metal plates of area S , at zero electrostatic potential and connected to each other. We shall show that the motion of an ion in the cell induces a current between these electrodes. This current oscillates at the angular frequency of the ion’s motion and can be measured.

To this end, we first need to prove a preliminary lemma, with no apparent connection to the problem under study.

- 37.** Consider a system of N point charges q_i located at positions $\vec{r}_i$, with $i = 0, \dots, N-1$. Write the expression of the electrostatic potential V_i created at the point $\vec{r}_i$ by the $N-1$ other charges q_j , with $j \neq i$.

- 38.** Now consider N charges with different values q'_i , but located at the same positions $\vec{r}_i$. Denote by V'_i the electrostatic potential created at $\vec{r}_i$ by the $N-1$ other charges q'_j . Prove the relation $\sum_{i=0}^{N-1} q_i V'_i = \sum_{i=0}^{N-1} q'_i V_i$, known as the reciprocity theorem.

We accept that this result remains valid if some of the point charges are replaced by electrodes of arbitrary shape whose positions remain fixed. Then q_i denotes the total charge carried by the electrode, and V_i its potential. In the case of interest, there are two electrodes at $y = -d/2$ and $y = d/2$, labelled by the indices $i = 1$ and $i = 2$. The point charge is labelled by the index $i = 0$; it is at $y = y_0$, with $-d/2 < y_0 < d/2$.

In the configuration of the experiment, the electrodes are at zero potential, so $V_1 = V_2 = 0$, and the point y_0 carries a charge q , so $q_0 = q$. We want to determine the charge difference between the two electrodes, $q_2 - q_1$.

- 39.** To this end, we consider a different charge distribution, in which there is no charge at y_0 , so $q'_0 = 0$, and the conductors are held at potentials $V'_1 = -V/2$ and $V'_2 = V/2$. Determine the potential V'_0 at the point y_0 .

- 40.** By applying the reciprocity theorem, deduce the charge difference $\Delta q = q_2 - q_1$ in terms of q , y_0 and d . Explain briefly how this device makes it possible to determine the resonance frequency.

41. For what practical reason are the detection electrodes placed perpendicular to the y axis rather than, for example, the x axis?

42. In practice, the excitation electric field does not oscillate at a single angular frequency but covers a whole range of frequencies. This makes it possible to analyse a range of values of m/z simultaneously, as in figure 2. Each frequency sets in motion the ions that are at resonance. After the end of the excitation, the measured variation of $\Delta q(t)$ with time t , which results from the motion of these ions, is also a superposition of a range of frequencies. The frequencies present can be reconstructed by spectral analysis, by digitising $\Delta q(t)$ at regular time intervals. Calculate the minimum sampling frequency needed to obtain the spectrum of figure 2 with a magnetic field $B = 7$ T. Given: $e/m_u = 10^8 \text{ C} \cdot \text{kg}^{-1}$, where m_u denotes the atomic mass unit.

43. From the point of view of detection, what is one further advantage of working with highly charged ions, such as those produced by electrospray, beyond the one already pointed out in question 21?

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